Search In Sorted Matrix

AlgoExpert · Medium · Searching

The problem

- Rows and columns are sorted ascending; return [row, col] of target or [-1, -1].
- Start top-right and eliminate a row or column each step.

Intuition

- Start at the top-right: moving left decreases, moving down increases, so each step eliminates a row or column.

Brute force vs optimal

- Scan every cell → O(r·c)
- Binary search each row → O(r log c)
- Start top-right, eliminate a row or column per step → O(r + c)

Approach

- 1. Start top-right
- 2. If too big, move left
- 3. If too small, move down
- 4. Return the coordinates or minus ones

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

```
1000 > 44 → move left
```

(0,5) = 1000

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

`15 < 44 → move down`

(0,4) = 15

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

`32 < 44 → move down`

(1,4) = 32

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

```
35 < 44 → move down
```

(2,4) = 35

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

```
45 > 44 → move left
```

(3,4) = 45

Worked example

1	4	7	12	15	1000
2	5	19	31	32	1001
3	8	24	33	35	1002
40	41	42	44	45	1003
99	100	103	106	128	1004

```
44 = 44 → found
```

(3,3) = 44

Java solution

```java
public class SearchInSortedMatrix {
    public static int[] searchInSortedMatrix(int[][] m, int target) {
        int r = 0, c = m[0].length - 1;
        while (r < m.length && c >= 0) {
            if (m[r][c] == target) return new int[]{r, c};
            if (m[r][c] > target) c--; else r++;
        }
        return new int[]{-1, -1};
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(1)

Edge cases & pitfalls

- Target absent → [−1, −1]
- Bottom-left start also works
- Top-left doesn't (both moves increase)

Interview tips & follow-ups

- Explain why the corner choice matters

Recap

- Start at the top-right: moving left decreases, moving down increases, so each step eliminates a row or column.
- Complexity: O(n + m)